

This version follows the approved survey's style: short, fully closed-ended, easy to chart, and each question produces one clear finding. FloodSense is introduced only after the existing problems have been measured.

FloodSense Resident Survey Questionnaire

Introduction / Panimula

We are studying how Bacoor City residents receive and use flood-related information. We want to understand what currently works, what problems residents experience, and whether a proposed FloodSense application could be useful.

Pinag-aaralan namin kung paano natatanggap at ginagamit ng mga residente ng Bacoor City ang impormasyon tungkol sa baha. Nais naming malaman kung ano ang kasalukuyang gumagana, anong mga problema ang nararanasan, at kung maaaring makatulong ang iminungkahing FloodSense application.

Participation is voluntary. Do not provide your name, exact address, or telephone number. Select only one answer unless the question says otherwise.

Kusang-loob ang pagsali. Huwag ibigay ang inyong pangalan, eksaktong tirahan, o numero ng telepono. Pumili lamang ng isang sagot maliban kung may ibang panuto.

Part I — Eligibility and Flood Experience

Q1. Are you 18 years old or older?

Kayo ba ay 18 taong gulang o mas matanda?

- Yes / Oo
- No / Hindi
- Prefer not to answer / Mas nais kong hindi sumagot

If No or Prefer not to answer, end the survey.

Q2. Do you currently live in a Bacoor City community included in this study?

Kasalukuyan ba kayong nakatira sa isang komunidad sa Bacoor City na kasama sa pag-aaral na ito?

- Yes / Oo
- No / Hindi

- I am not sure / Hindi ako tiyak

If No, end the survey.

Q3. During the past three years, how often has flooding or heavy rain directly affected your household, travel, work, or studies?

Sa nakalipas na tatlong taon, gaano kadalas direktang naapektuhan ng baha o malakas na ulan ang inyong sambahayan, pagbiyahe, trabaho, o pag-aaral?

- Very often / Napakadalas
- Often / Madalas
- Sometimes / Paminsan-minsan
- Rarely / Bihira
- Never / Hindi kailanman
- I cannot recall / Hindi ko maalala

Part II — Current Flood-Information Experience

Q4. Which source do you usually check first for flood-related information?

Aling pinagmumulan ang karaniwan ninyong unang tinitingnan para sa impormasyon tungkol sa baha?

- Official government social-media page / Opisyal na social-media page ng pamahalaan
- Barangay announcement or public-address system / Anunsiyo o public-address system ng barangay
- Text or SMS alert / Text o SMS alert
- Television or radio / Telebisyon o radyo
- Government or weather website / Website ng pamahalaan o serbisyong pangpanahon
- Family, neighbors, or community group / Pamilya, kapitbahay, o grupong pangkomunidad
- Personal observation / Sariling pagmamasid

- I do not usually check any source / Wala akong karaniwang tinitingnang pinagmulan

Q5. During flooding or heavy rain, how often are you able to obtain the information you need?

Kapag may baha o malakas na ulan, gaano kadalas ninyong nakukuha ang impormasyong kailangan ninyo?

- Always / Palagi
- Often / Madalas
- Sometimes / Paminsan-minsan
- Rarely / Bihira
- Never / Hindi kailanman
- I usually do not need flood information / Karaniwan ay hindi ko kailangan ang impormasyon

Q6. When does useful flood information usually reach you?

Kailan karaniwang dumarating sa inyo ang kapaki-pakinabang na impormasyon tungkol sa baha?

- Early enough to prepare or decide / Sapat ang aga upang makapaghandang o makapagdesisyon
- Shortly before I need to act / Kaunting oras bago ko kailangang kumilos
- After I have already started acting / Pagkatapos kong magsimulang kumilos
- After flooding has already affected me / Pagkatapos akong maapektuhan ng baha
- I usually receive no useful information / Karaniwan ay wala akong natatanggap na kapaki-pakinabang na impormasyon
- I cannot recall / Hindi ko maalala

Q7. How easy or difficult is the flood information you receive to understand?

Gaano kadali o kahirap unawain ang impormasyong natatanggap ninyo tungkol sa baha?

- Very easy / Napakadali
- Easy / Madali
- Neither easy nor difficult / Hindi madali at hindi rin mahirap

- Difficult / Mahirap
- Very difficult / Napakahirap
- I do not receive flood information / Wala akong natatanggap na impormasyon

Q8. How well does the information identify the location that may be affected?

Gaano kahusay tinutukoy ng impormasyon ang lugar na maaaring maapektuhan?

- It gives all the location details I need / Ibinibigay nito ang lahat ng detalyeng kailangan ko
- It gives some, but not all, location details / May ibinibigay itong detalye ngunit hindi kumpleto
- It is too general to help me identify the affected location / Masyado itong pangkalahatan upang matukoy ang apektadong lugar
- Location details are not important for my decisions / Hindi mahalaga sa aking desisyon ang detalye ng lugar
- I do not receive flood information / Wala akong natatanggap na impormasyon
- I am not sure / Hindi ako tiyak

Q9. How often do different flood-information sources provide conflicting or inconsistent information?

Gaano kadalas nagbibigay ng magkakasalungat o hindi magkakatugmang impormasyon ang iba't ibang pinagmumulan?

- Very often / Napakadalas
- Often / Madalas
- Sometimes / Paminsan-minsan
- Rarely / Bihira
- Never / Hindi kailanman
- I use only one source / Isang pinagmumulan lamang ang ginagamit ko
- I am not sure / Hindi ako tiyak

Q10. Which statement best describes how the available information affects your decisions?

Aling pahayag ang pinakamahasag na naglalarawan kung paano nakaaapekto sa inyong mga desisyon ang impormasyong mayroon kayo?

- It usually helps me decide in time / Karaniwan ay nakatutulong ito upang makapagdesisyon ako sa tamang oras
- It helps, but sometimes arrives too late / Nakatutulong ito ngunit kung minsan ay huli nang dumarating
- Information is available but is not useful for my decision / May impormasyon ngunit hindi ito kapaki-pakinabang sa aking desisyon
- I usually cannot obtain the information I need / Karaniwan ay hindi ko nakukuha ang impormasyong kailangan ko
- I usually do not need information to decide / Karaniwan ay hindi ko kailangan ng impormasyon upang makapagdesisyon
- I am not sure / Hindi ako tiyak

Q11. What is the most common problem you experience with flood information?

Ano ang pinakakaraniwang problemang nararanasan ninyo sa impormasyon tungkol sa baha?

- It arrives too late / Huli itong dumarating
- It is difficult to understand / Mahirap itong unawain
- It does not identify the affected location clearly / Hindi nito malinaw na tinutukoy ang apektadong lugar
- Different sources provide inconsistent information / Hindi magkakatatugma ang impormasyon mula sa iba't ibang pinagmumulan
- I cannot access it when needed / Hindi ko ito ma-access kapag kailangan
- It appears outdated / Tila luma na ang impormasyon
- I do not usually experience a problem / Karaniwan ay wala akong nararanasang problema
- I am not sure / Hindi ako tiyak

Q12. For planning before heavy rain, how far in advance is flood-related information most useful to you?

Para sa paghahanda bago ang malakas na ulan, gaano kaagang impormasyon tungkol sa baha ang pinakakapaki-pakinabang sa inyo?

- Less than 30 minutes before / Wala pang 30 minuto bago
- About 30 minutes to 1 hour before / Humigit-kumulang 30 minuto hanggang 1 oras bago
- About 2 to 3 hours before / Humigit-kumulang 2 hanggang 3 oras bago
- About 4 to 6 hours before / Humigit-kumulang 4 hanggang 6 na oras bago
- More than 6 hours before / Mahigit 6 na oras bago
- It depends on the situation / Depende sa sitwasyon
- I do not need advance information / Hindi ko kailangan ng maagang impormasyon

Q13. Which type of flood information is most important to you?

Aling uri ng impormasyon tungkol sa baha ang pinakamahalaga sa inyo?

- Current official warning or advisory / Kasalukuyang opisyal na babala o abiso
- Whether my area may be susceptible to flooding / Kung maaaring bahain ang aking lugar
- Current street or road conditions / Kasalukuyang kalagayan ng kalsada o lansangan
- Expected or observed water depth / Inaasahan o naobserbahang lalim ng tubig
- Routes officially reported as open or closed / Mga rutang opisyal na iniulat na bukas o sarado
- Verified evacuation-center information / Beripikadong impormasyon tungkol sa evacuation center
- Household preparedness instructions / Mga tagubilin sa paghahanda ng sambahayan
- Emergency contact information / Impormasyon sa emergency contact
- I do not need additional information / Hindi ko kailangan ng dagdag na impormasyon

Q14. Overall, how well do your current sources meet your flood-information needs?

Sa kabuuan, gaano kahusay natutugunan ng inyong kasalukuyang mga pinagmumulan ang pangangailangan ninyo sa impormasyon tungkol sa baha?

- Completely / Ganap
 - Mostly / Karamihan
 - Partly / Bahagya
 - Very little / Kaunti lamang
 - Not at all / Hindi talaga
 - I do not need flood information / Hindi ko kailangan ng impormasyon tungkol sa baha
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Part III — Technology Access and Preference

Q15. Which statement best describes your usual access to a mobile phone?

Aling pahayag ang pinakamahusay na naglalarawan sa karaniwan ninyong access sa mobile phone?

- I regularly use my own smartphone / Regular kong ginagamit ang sarili kong smartphone
- I regularly use a shared smartphone / Regular kong ginagamit ang smartphone na pinagsasaluhan
- I use a basic phone without applications / Basic phone na walang applications ang ginagamit ko
- I do not regularly have access to a mobile phone / Wala akong regular na access sa mobile phone
- Prefer not to answer / Mas nais kong hindi sumagot

Q16. During heavy rain, how reliable is your mobile-data or internet access?

Kapag malakas ang ulan, gaano maaasahan ang inyong mobile data o internet access?

- Always reliable / Palaging maaasahan
- Usually reliable / Kadalasang maaasahan
- Sometimes reliable / Paminsan-minsang maaasahan
- Rarely reliable / Bihirang maaasahan
- Never reliable / Hindi kailanman maaasahan

- I do not use mobile data or internet / Hindi ako gumagamit ng mobile data o internet
- I am not sure / Hindi ako tiyak

Q17. How would you most prefer to receive flood information?

Sa anong paraan ninyo pinakanais matanggap ang impormasyon tungkol sa baha?

- Text or SMS / Text o SMS
- Official social media / Opisyal na social media
- Barangay public-address announcement / Public-address announcement ng barangay
- Television or radio / Telebisyon o radyo
- Mobile application / Mobile application
- Government website / Website ng pamahalaan
- Telephone hotline / Telephone hotline
- Family or community network / Pamilya o network ng komunidad
- I have no particular preference / Wala akong partikular na gusto
- I do not want additional information / Ayaw ko ng dagdag na impormasyon

Part IV — Perception of the Proposed FloodSense Concept

Concept Description / Paglalarawan ng Konsepto

FloodSense is a proposed Android application for planning before possible flooding in Bacoor City. A user would select a supported rainfall-intensity category, rainfall duration, and location. Using approved geographic information and rules reviewed by experts, the application would display a flood-susceptibility category and an explanation. It may also provide verified evacuation-center records and preparedness guidance.

FloodSense is not a live flood forecast, a live street-monitoring service, a guarantee that a route is safe, or a replacement for official government warnings.

Ang FloodSense ay isang iminungkahing Android application para sa paghahanda bago ang posibleng pagbaha sa Bacoor City. Pipili ang user ng suportadong kategorya ng lakas ng ulan, tagal ng pag-ulan, at lokasyon. Gamit ang aprubadong impormasyong heograpiko at mga patakarang sinuri ng eksperto, magpapakita ang application ng antas ng flood

susceptibility at paliwanag. Maaari rin itong magbigay ng beripikadong rekord ng evacuation center at gabay sa paghahanda.

Ang FloodSense ay hindi live flood forecast, live na pagmamanman sa bawat kalsada, garantiya ng ligtas na ruta, o kapalit ng opisyal na babala ng pamahalaan.

For Questions 18–22, use this scale:

Para sa Tanong 18–22, gamitin ang sumusunod na scale:

- Very useful / Lubhang kapaki-pakinabang
- Useful / Kapaki-pakinabang
- Neither useful nor unhelpful / Hindi kapaki-pakinabang at hindi rin walang pakinabang
- Slightly useful / Bahagyang kapaki-pakinabang
- Not useful / Hindi kapaki-pakinabang
- I would not use this feature / Hindi ko gagamitin ang feature na ito
- I am not sure / Hindi ako tiyak

Q18. How useful would selecting a rainfall-intensity and duration scenario be when planning before possible flooding?

Gaano magiging kapaki-pakinabang ang pagpili ng sitwasyon ng lakas at tagal ng pag-ulan sa paghahanda bago ang posibleng pagbaha?

Q19. How useful would a map showing the flood-susceptibility category of a supported area be?

Gaano magiging kapaki-pakinabang ang mapang nagpapakita ng antas ng flood susceptibility ng isang suportadong lugar?

Q20. How useful would an explanation of the susceptibility result and the date of the information be?

Gaano magiging kapaki-pakinabang ang paliwanag sa susceptibility result at petsa ng impormasyong ginamit?

Q21. How useful would verified evacuation-center locations and available status information be?

Gaano magiging kapaki-pakinabang ang beripikadong lokasyon ng evacuation center at impormasyong mayroon tungkol sa kalagayan nito?

Q22. How useful would household preparedness guidance based on the displayed situation be?

Gaano magiging kapaki-pakinabang ang gabay sa paghahanda ng sambahayan batay sa ipinakitang sitwasyon?

Q23. How likely would you be to use FloodSense for planning before possible flooding?

Gaano kalamang na gagamitin ninyo ang FloodSense para sa paghahanda bago ang posibleng pagbaha?

- Very likely / Lubhang malamang
- Likely / Malamang
- Neither likely nor unlikely / Katamtaman ang posibilidad
- Unlikely / Hindi malamang
- Very unlikely / Lubhang hindi malamang
- I am not sure / Hindi ako tiyak

Q24. What would be your main concern about using FloodSense?

Ano ang magiging pangunahing alalahanin ninyo sa paggamit ng FloodSense?

- Information might be inaccurate / Maaaring hindi tama ang impormasyon
- Information might be outdated / Maaaring luma na ang impormasyon
- Internet access might be unavailable / Maaaring walang internet access
- The application might be difficult to understand / Maaaring mahirap unawain ang application
- The application might use too much mobile data or battery / Maaaring malakas gumamit ng mobile data o baterya
- I might depend on it instead of official warnings / Maaaring umasa ako rito sa halip na sa opisyal na babala
- I would have no major concern / Wala akong malaking alalahanin

- I would not use FloodSense / Hindi ko gagamitin ang FloodSense
- I am not sure / Hindi ako tiyak

Q25. What role should FloodSense have in the way you receive flood information?

Anong papel ang dapat gampanan ng FloodSense sa paraan ng pagtanggap ninyo ng impormasyon tungkol sa baha?

- It should supplement official government warnings and my current sources / Dapat itong pandagdag sa opisyal na babala at kasalukuyan kong mga pinagmumulan
- It should be used mainly for advance planning / Dapat itong gamitin pangunahin para sa maagang paghahanda
- It should be used only when I need additional information / Dapat lamang itong gamitin kapag kailangan ko ng dagdag na impormasyon
- I prefer to continue using my current sources without FloodSense / Mas nais kong ipagpatuloy ang kasalukuyan kong mga pinagmumulan nang walang FloodSense
- I do not need flood-related information / Hindi ko kailangan ng impormasyon tungkol sa baha
- I am not sure / Hindi ako tiyak

Researcher Interpretation Guide

Q1–Q2: Eligibility

Use these questions only to determine eligibility. Do not treat excluded residents as negative responses.

Q3: Flood exposure frequency

Shows how frequently the surveyed residents have been affected.

Suggested interpretation:

Among the surveyed residents, ___% reported being affected often or very often during the past three years, while ___% reported rare or no direct effects.

Do not claim that the percentage represents all Bacoar residents unless the sampling design supports that conclusion.

Q4: First information source

Identifies the source residents depend on first. This can guide channel integration and communication partnerships.

Suggested interpretation:

The most frequently selected first source was _____, chosen by ____% of respondents.

It does not establish that the selected source is accurate or effective.

Q5: Information availability

Measures how consistently residents can obtain needed information.

Suggested interpretation:

A total of ____% reported that they could always or often obtain the information they needed, while ____% reported that they could rarely or never obtain it.

Q6: Information timing

Determines whether information reaches residents before or after they need to act.

Suggested interpretation:

____% received useful information early enough to prepare or decide, whereas ____% received it only after acting or after flooding had affected them.

This reports timing; it does not prove why information was delayed.

Q7: Understandability

Measures perceived clarity.

Suggested interpretation:

____% described the information as easy or very easy to understand, while ____% described it as difficult or very difficult.

Q8: Location detail

Determines whether current information gives residents sufficient geographic detail.

Suggested interpretation:

____% reported receiving all the location detail they needed, while ____% considered the available information too general.

This does not establish the accuracy of official maps.

Q9: Inconsistency

Measures how often residents perceive conflicting information.

Suggested interpretation:

___% reported encountering inconsistent information often or very often.

Report this as respondent perception unless the conflicting messages are independently verified.

Q10: Decision usefulness

Shows whether current information supports timely decisions.

Suggested interpretation:

___% reported that available information usually helped them decide in time. However, ___% said it arrived too late, was not useful, or could not be obtained.

Q11: Main information problem

Identifies the most common single problem and prevents the analysis from treating every respondent as having the same difficulty.

Suggested interpretation:

The most frequently identified problem was _____ at ____%. Meanwhile, ___% reported that they did not usually experience a problem.

Q12: Preferred advance-information period

Identifies when information is most useful for planning.

Suggested interpretation:

The most frequently selected planning period was _____.

Do not interpret the answer as a promise that FloodSense can provide a forecast that far in advance.

Q13: Highest information priority

Determines which type of information should receive the greatest research or design attention.

Suggested interpretation:

_____ was the most frequently selected information priority, followed by _____.

Selecting a priority does not prove that FloodSense can safely or legally provide it.

Q14: Adequacy of current methods

Measures whether existing sources already meet residents' needs.

Suggested interpretation:

___% said their current sources completely or mostly met their needs, while ___% said their needs were met very little or not at all.

This question is important because it permits a finding that no replacement is needed.

Q15–Q17: Technology feasibility

These questions measure smartphone access, internet reliability, and preferred communication channel.

Suggested interpretation:

Although ___% had regular smartphone access, only ___% reported usually or always reliable connectivity during heavy rain. The most preferred communication channel was _____.

Smartphone access alone does not establish willingness or ability to use FloodSense during an emergency.

Q18–Q22: Feature usefulness

Calculate the percentage selecting Useful or Very useful for each component. Rank the components from highest to lowest.

Suggested interpretation:

The highest-rated component was _____, with ___% rating it useful or very useful. The lowest-rated component was _____ at ___%.

These questions measure perceived usefulness, not actual effectiveness.

Q23: Intention to use

Measures stated likelihood of using the proposed concept.

Suggested interpretation:

___% reported that they were likely or very likely to use FloodSense for pre-event planning.

Do not report intended use as proof of actual future adoption.

Q24: Main concern

Identifies the most important adoption or safety barrier.

Suggested interpretation:

The most frequently selected concern was _____, reported by ____% of respondents. Meanwhile, ____% reported no major concern.

Q25: Appropriate role of FloodSense

Determines whether residents view FloodSense as a supplement, planning tool, occasional source, or unnecessary option.

Suggested interpretation:

Most respondents preferred FloodSense to function as _____. Only ____% preferred to continue without the proposed application.

This question must guide the project's claims. FloodSense should not be presented as a replacement for official warnings unless that role has been formally authorized and validated.